Label Consistency and Iconicity in Tactile Category Learning

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Objectives

- ❖ Linguistic labels shape visual categorization (e.g., Lupyan et al., 2007), but their role in tactile perception (a modality difficult to verbalize (Majid & Levinson, 2011)) remains unexplored.
- ❖ Japanese mimetic words offer iconic sound-meaning correspondences (Imai & Kita, 2014).
- ❖ The Label Feedback Hypothesis predicts: Labels provide top-down feedback enhancing perceptual discrimination (Lupyan, 2012).

Building on perceptual learning research in tactile and olfactory domains (Miller et al., 2018 ; Vanek et al., 2021),

RQ: Does label-percept consistency facilitate tactile category learning, and do effects reflect genuine perceptual reorganization or label-dependent associations?

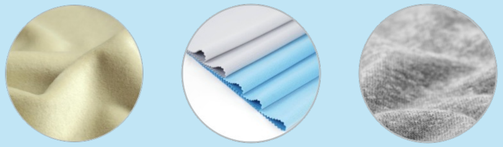
H1: Consistent label-texture pairings produce larger learning gains than inconsistent pairings.

H2: Effects persist without labels, indicating genuine category formation rather than label-dependent associations.

Experiment : Consistency Effect on Tactile Category Learning

Participants: 52 Japanese speakers (N = 26 per condition)

Tactile Stimuli: 18 fabric types selected from Hanada (2023), organized by friction, softness, and thickness.



Fleece | Microfiber | Jersey

Example triplet: similar fabrics grouped by softness / thickness (Hanada, 2023)

CONCORDANT
75% JP-Match / 25% Pseudo

- 12/16 mokomoko, 4/16 senoseno, mootite..
- 12/16 huwahuwa, 4/16 mupomupo, dastude..
- 12/16 mohumohu, 4/16 himohimo, bitjeed..

DISCORDANT
25% JP-Match / 75% Pseudo

- 4/16 mokomoko, 12/16 senoseno, mootite..
- 4/16 huwahuwa, 12/16 mupomupo, dastude..
- 4/16 mohumohu, 12/16 himohimo, bitjeed..

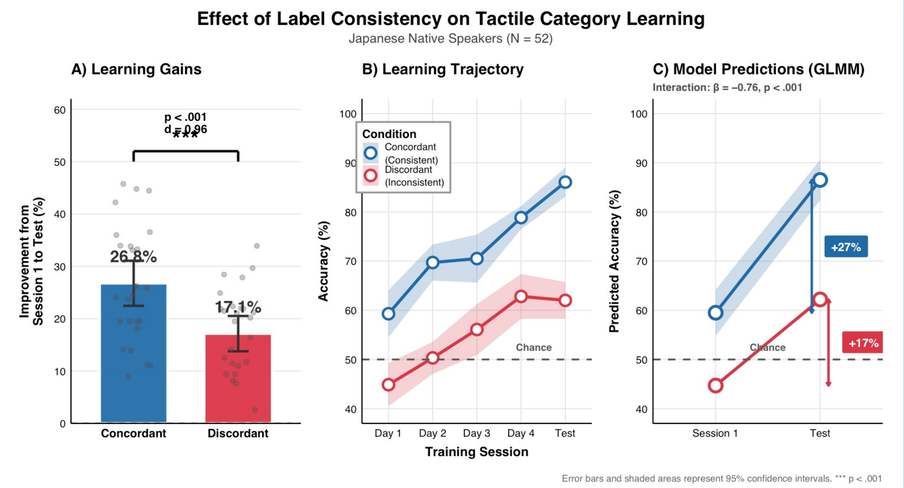
Label A (Tactile A (6s)) → Label B (Tactile B (6s)) → Label X (Tactile X (6s)) → A or B? Confidence 1-9, ✅ / ❌

Unlabeled Test
No labels • Day 1 triplets • Pure perceptual

4 blocks x 6 triplets/block × 4 days - Unlabeled test (post Day 4)

Type	Structure	Examples
JP-Match	AB-AB, Sound-symbolic, mimetic Selected from Hanada (2023); National Language Database	Mokomoko, Huwahuwa, Mohumohu
JP-Pseudo	AB-AB, Non-sound symbolic, JP mimetic-like Novel pseudowords using vowel/consonant pairs absent in Japanese	Mupomupo, Senoseno, Himohimo
ENG-Pseudo	Bisyllabic pseudowords, arbitrary Selected from Rastle & Coltheart (2000)	Mootie, Dastude, Bitjeed

Results



Accuracy: Learning Gains
Model: accuracy ~ condition × session_type + (1|participant) + (1|triplet_id)

Learning Trajectory: **Concordant**: Smooth progression
Discordant: Struggled throughout

- Gap emerged early and persisted
- Concordant participants maintained 86% accuracy at unlabeled test
- Consistent labels amplified learning by 54% over inconsistent labels.

Reaction Time
Model: rt_decision ~ condition × accuracy + (1|participant) + (1|triplet_id)

- 17% speed advantage
- Speed benefit applied equally to correct/incorrect responses
- Enhanced processing fluency, not speed-accuracy tradeoff

Confidence Ratings

Concordant: Confidence stable (Δ = -0.07), difference: p < .001
Confidence tracked actual learning (r = .37, p = .059)

Discordant: Confidence increased despite minimal learning gains (Δ = +0.65)
Confidence decoupled from learning (r = .11, p = .60)

- Inconsistent labels may produce illusory confidence, participants feel they are learning despite minimal actual improvement.

Conclusions

Consistent iconic labels amplified tactile category learning by 54%, with effects persisting at unlabeled test

❖ **Genuine perceptual reorganization, not label-dependent associations.**

Extending the Label Feedback Hypothesis to Touch

- ❖ Consistent iconic labels
  → Top-down feedback directing attention to category-relevant features (friction, softness, thickness)
- ❖ Repeated consistent pairing
  → Reshapes perceptual space, enhancing within-category similarity and between-category distinctiveness (Goldstone, 1998)
- ❖ Misaligned labels
  → Existing sound-symbolic knowledge creates conflicting top-down effects, actively hindering learning
- ❖ Effects persist without labels
  → categories internalized, not merely memorized

🔑 **Consistency > Iconicity Per Se**

- ❖ Native Japanese speakers possess rich mimetic vocabularies, yet showed robust sensitivity to label-percept consistency
  - Iconicity did not universally facilitate learning; it created interference when labels misaligned
  - Label-percept consistency matters more than iconicity alone

- ❖ **Labels shape perception in touch, not only vision/audition extending linguistic relativity and the LFH across sensory modalities**
- ❖ **Consistent, meaningful labels can scaffold category formation in tactile discrimination domains**
- ❖ **Existing linguistic knowledge both helps (when consistent) and hinders (when inconsistent) perceptual learning**

Selected References: Akita, K. (2009). A grammar of sound- symbolic words in Japanese. Doctoral dissertation, Kobe University.; Hanada, M. (2023). Dimensional structure of tactile textures characterized by Japanese onomatopoeia. *Attention, Perception, & Psychophysics*, 85, 2669-2680.; Imai, M., & Kita, S. (2014). The sound symbolism bootstrapping hypothesis for language acquisition and evolution. *Philosophical Transactions of the Royal Society B*, 369(1651), 20130298.; Lupyan, G. (2012). Linguistically modulated perception and cognition: The label-feedback hypothesis. *Frontiers in Psychology*, 3, 54.; Vanek, N., Sóskuthy, M., & Majid, A. (2021). Consistent verbal labels promote odor category learning. *Cognition*, 206, 104485.